INF10025 Data Management and Analytics

Task 3 – Credit Submission

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A. Task Credit3a

Firstly, the first task of the third laboratory in the Credit section requires students to write a query that meets the following requirements:

- For each row in the MOVIE table, display the stuid, Movieno, title, release year, ratingcode, runtime, the matching short description from the RATING table and the colour name from the COLOURTYPE table.
- Only display rows that have runtime of greater than 170 minutes .
- Only display rows that have a rating code of M or MA.
- The list must be in Ascending Movie no sequence.

Enter Statements:

```
SELECT '105565520' AS StuID, M.Movieno, M.Title, M.RELYEAR, M.RATINGCODE,
M.Runtime, R.SHORTDESC, C.COLOURNAME
FROM MOVIE5520 M
INNER JOIN RATING5520 R ON M.RATINGCODE = R.RATINGCODE
INNER JOIN COLOURTYPE5520 C ON M.COLOURCODE = C.COLOURCODE
WHERE M.RUNTIME > 170
AND (M.RATINGCODE = 'M' OR M.RATINGCODE='MA')
ORDER BY M.Movieno ASC;
```

Figure 1: SQL Query Statement for Task Credit3a.

STUID	MOVIENO	TITLE	RELYEAR	RATINGCODE	RUNTIME	SHORTDESC	COLOURNAME
105565520	120	The Lord of the Rings: The Fellowship of the Ring	2001	M	178	MATURE (15+)	Colour Movie
105565520	121	The Lord of the Rings: The Two Towers	2002	M	179	MATURE (15+)	Colour Movie
105565520	122	The Lord of the Rings: The Return of the King	2003	M	201	MATURE (15+)	Colour Movie
105565520	424	Schindlers List	1993	M	195	MATURE (15+)	Black & White Movie
105565520	597	Titanic	1997	M	194	MATURE (15+)	Colour Movie
105565520	676	Pearl Harbor	2001	M	183	MATURE (15+)	Colour Movie
105565520	10549	Hamlet	1996	M	242	MATURE (15+)	Colour Movie
105565520	83542	Cloud Atlas	2012	MA	172	MATURE ACCOMPANIED (15+)	Colour Movie
105565520	398978	The Irishman	2019	MA	209	MATURE ACCOMPANIED (15+)	Colour Movie

Figure 2: Output Result for Task Credit3a SQL Query.

B. Task Credit3b

Next, for the Credit 3b task, students are required to write a SQL statement that does the following:

- Display the StuId, MovieNo, Title, Runtime, Rating code, Rating Short Description, tmdb score for movies that meet any of these criteria:
- Rating code of MA plus a runtime between 100-105 (inclusive)
- Rating code of G plus a runtime less than 90
- Rating code of PG plus a runtime between 120-125 (inclusive)
- Rating code of M plus a runtime 170 minutes or more
- The query must only include movies that have a tmdb_score between 6.8 and 7.4 inclusive
- The list must be in Ascending Movie no sequence.

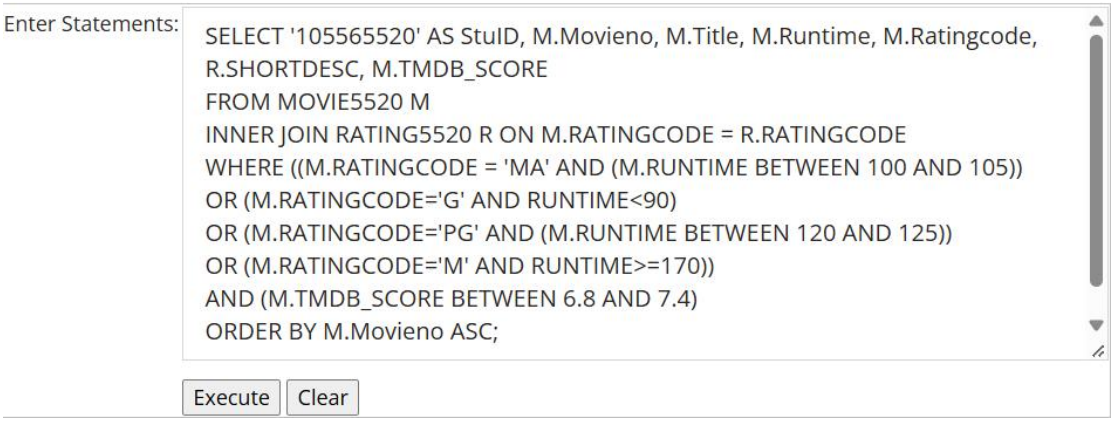


Figure 3: SQL Query Statement for Task Credit3b.

STUID	MOVIENO	TITLE	RUNTIME	RATINGCODE	SHORTDESC	TMDB_SCORE
105565520	153	Lost in Translation	102	MA	MATURE ACCOMPANIED (15+)	7.4
105565520	557	Spider-Man	121	PG	PARENTAL GUIDANCE	7
105565520	676	Pearl Harbor	183	M	MATURE (15+)	6.8
105565520	2567	The Aviator	170	M	MATURE (15+)	7.1
105565520	2759	The Adventures of Priscilla, Queen of the Desert	104	MA	MATURE ACCOMPANIED (15+)	7.3
105565520	18785	The Hangover	100	MA	MATURE ACCOMPANIED (15+)	7.2
105565520	257211	The Intern	121	PG	PARENTAL GUIDANCE	7.1
105565520	416477	The Big Sick	120	PG	PARENTAL GUIDANCE	7.4
105565520	419704	Ad Astra	123	PG	PARENTAL GUIDANCE	7

Figure 4: Output Result for Task Credit3b SQL Query.

C. Task Credit3c

Next, in the third task of the Credit section, the laboratory requires students to write a single SQL statement that ***displays all the release years in the movie table***. Furthermore, the ***list must not repeat any value***, and the ***resulting dataset must be sorted in descending order by release year.***

Enter Statements:

```
SELECT DISTINCT '105565520' AS StudID, RELYEAR
FROM MOVIE5520
ORDER BY RELYEAR DESC;
```

Figure 5: SQL Query Statement for Task Credit3c.

STUDID	RELYEAR
105565520	2019
105565520	2018
105565520	2017
105565520	2016
105565520	2015
105565520	2014
105565520	2013
105565520	2012
105565520	2011
105565520	2010
105565520	2009
105565520	2008
105565520	2007
105565520	2006

Figure 6: Output Result for Task Credit3c SQL Query.

D. Task Credit3d

In the fourth task of the Credit section, the laboratory requires updating specific movie records based on runtime and rating conditions, verifying the changes, and displaying the filtered datasets according to defined sorting criteria.

Enter Statements:

```
UPDATE Movie5520
SET TMDB_VOTES = 0;
```

Execute Clear

OK. 395 rows

Figure 7: Update Statement to Set TMDB_VOTES to 0.

Enter Statements:

```
SELECT * FROM MOVIE5520
ORDER BY Movieno DESC;
```

Figure 8: SQL Query Statement to Display Movies in Descending Movieno Sequence.

MOVIENO	TITLE	RELYEAR	RUNTIME	RATINGCODE	COLOURCODE	TMDB_SCORE	TMDB_VOTES	IMDB_ID
549053	Last Christmas	2019	102	PG	C	7.2	0	tt8623904
512200	Jumanji: The Next Level	2019	98	PG	C	7.3	0	tt7975244
489925	Eighth Grade	2018	94	M	C	7.6	0	tt7014006
475557	Joker	2019	122	MA	C	8.8	0	tt7286456
466272	Once Upon a Time... in Hollywood	2019	161	MA	C	8	0	tt7131622
459992	Long Shot	2019	125	MA	C	6.9	0	tt2139881
458423	Mamma Mia! Here We Go Again	2018	113	PG	C	7.3	0	tt6911608
454983	The Kissing Booth	2018	105	PG	C	7.3	0	tt3799232
445571	Game Night	2018	101	M	C	6.9	0	tt2704998
433310	When We First Met	2018	97	PG	C	6.6	0	tt5783956
419704	Ad Astra	2019	123	PG	C	7	0	tt2935510

Figure 9: Output Result Showing TMDB_VOTES Set to 0.

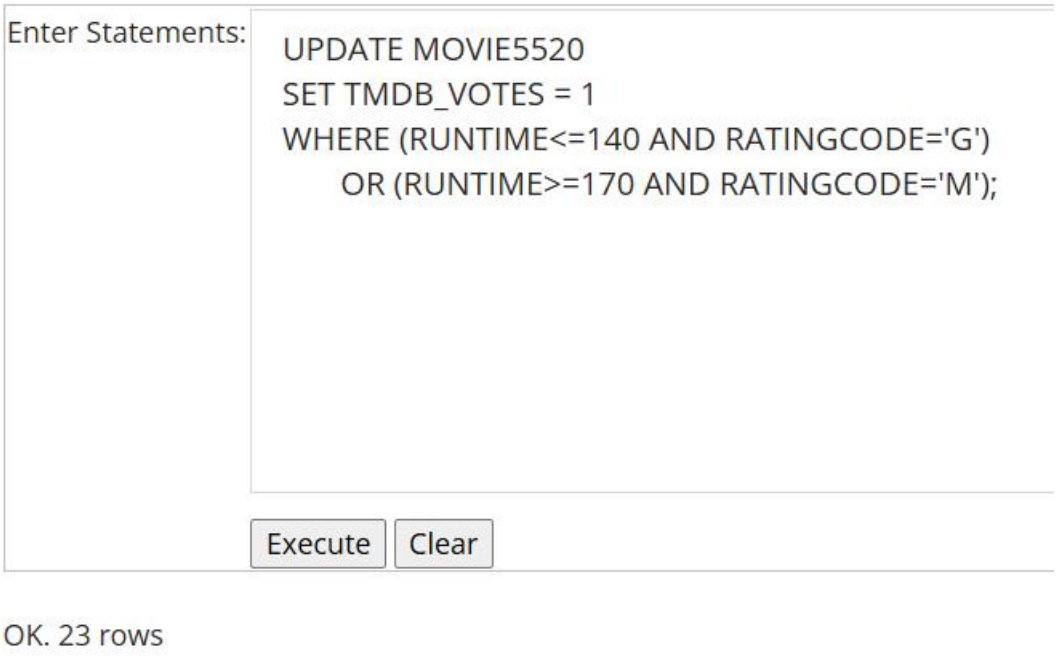


Figure 10: Update Statement to Set Conditional Values.

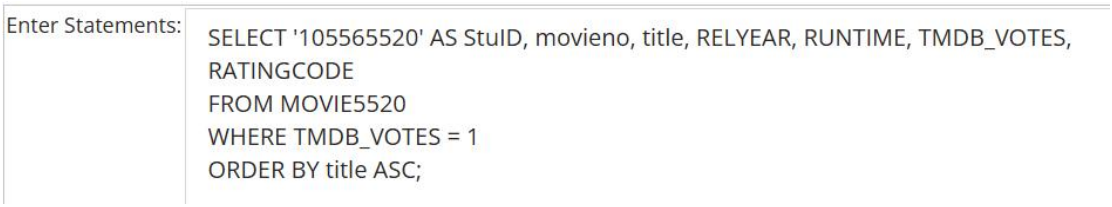


Figure 11: SQL Query Statement for Movies with TMDB_VOTES Equal to 1.

STUID	MOVIENO	TITLE	RELYEAR	RUNTIME	TMDB_VOTES	RATINGCODE
105565520	218778	Alexander and the Terrible, Horrible, No Good, Very Bad Day	2014	81	1	G
105565520	127380	Finding Dory	2016	103	1	G
105565520	16084	Freaky Friday	1976	95	1	G
105565520	109445	Frozen	2013	102	1	G
105565520	45772	Gnomeo & Juliet	2011	84	1	G
105565520	10549	Hamlet	1996	242	1	M
105565520	9836	Happy Feet	2006	108	1	G
105565520	76492	Hotel Transylvania	2012	91	1	G
105565520	62211	Monsters University	2013	104	1	G
105565520	585	Monsters, Inc.	2001	92	1	G

Figure 12: Output Result for Movies with TMDB_VOTES Equal to 1.

E. Task Credit3e

In the task Credit3e, the laboratory requires students to write an SQL statement that displays the ***student ID*** alongside all columns from the movie table. In addition, the query must filter the results to ***only include movies whose titles do not contain any of the letters A, B, I, or O***, regardless of case. Finally, the resulting list must be ***sorted in descending order by the movie title.***

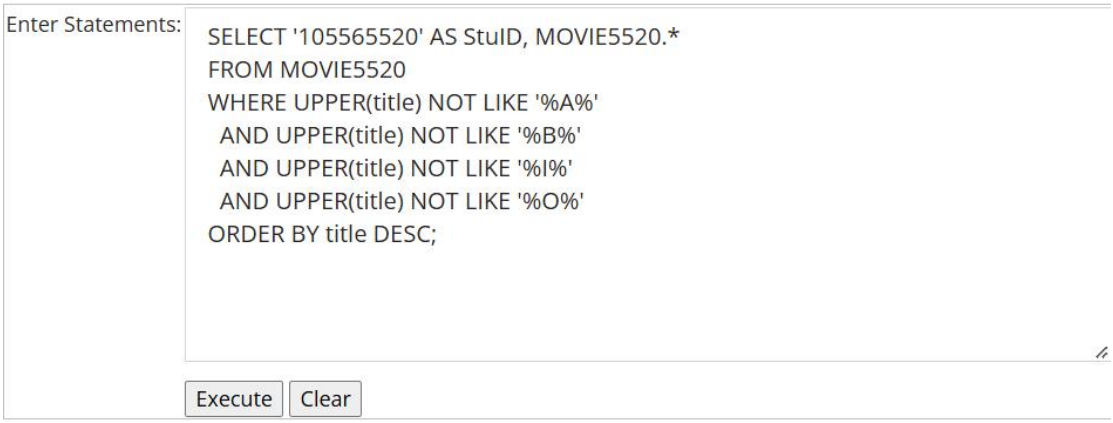


Figure 13: SQL Query Statement for Task Credit3e.

STUID	MOVIENO	TITLE	RELYEAR	RUNTIME	RATINGCODE	COLOURCODE	TMDB_SCORE	TMDB_VOTES	IMDB_ID
105565520	36657	X-Men	2000	104	M	C	6.9	0	tt0120903
105565520	564	The Mummy	1999	124	M	C	6.7	0	tt0120616
105565520	72105	Ted	2012	106	MA	C	6.3	0	tt1637725
105565520	23629	Sucker Punch	2011	110	MA	C	6.1	0	tt0978764
105565520	238713	Spy	2015	120	M	C	6.8	0	tt3079380
105565520	1637	Speed	1994	116	MA	C	6.9	0	tt0111257
105565520	206647	Spectre	2015	148	M	C	6.2	0	tt2379713
105565520	807	Se7en	1995	127	MA	C	8.3	0	tt0114369
105565520	9824	Mystery Men	1999	121	PG	C	5.8	0	tt0132347
105565520	2022	Mr. Deeds	2002	96	M	C	5.7	0	tt0280590
105565520	10710	Envy	2004	99	PG	C	4.8	0	tt0326856
105565520	212778	Chef	2014	114	PG	C	7.1	0	tt2883512

Figure 14: Output Result for Task Credit3e.

F. Task Credit3f

In this section, students have to create a table named ***CAR5520*** based on the relational schema and the ERD below. Then, students have to save the SQL script into a file named ***create_car_sql***.txt.

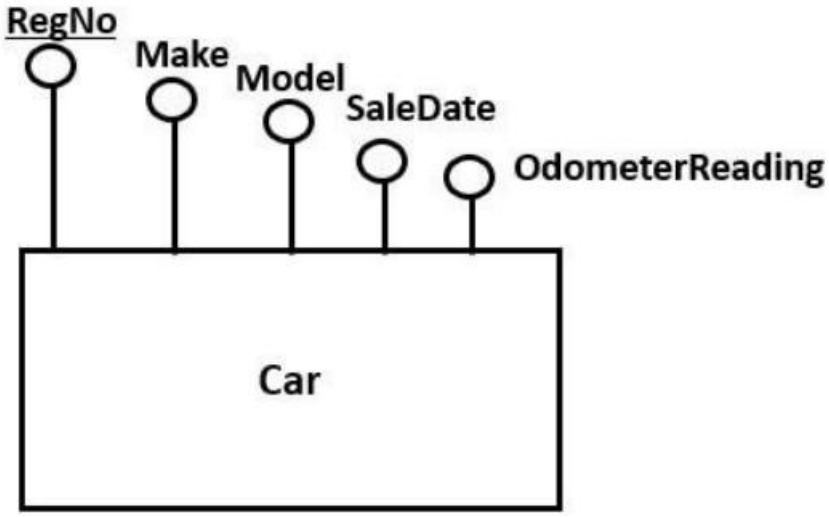


Figure 15: Entity-Relationship Diagram for Car Entity.

create_car_sql - Notepad

File Edit View

```
CREATE TABLE CAR5520 (
  RegNo            VARCHAR(50) PRIMARY KEY,
  Make             VARCHAR(50) NOT NULL,
  Model            VARCHAR(50) NOT NULL,
  SaleDate         DATE NOT NULL,
  OdometerReading  NUMBER NOT NULL
);
```

Figure 16: SQL Script to Create CAR5520 Table.

Enter Statements:

```
CREATE TABLE CAR5520 (
  RegNo            VARCHAR(50) PRIMARY KEY,
  Make             VARCHAR(50) NOT NULL,
  Model            VARCHAR(50) NOT NULL,
  SaleDate         DATE NOT NULL,
  OdometerReading  NUMBER NOT NULL
);
```

Execute Clear

OK. 0 rows

Figure 17: Execution Result for Creating CAR5520 Table.

G. Task Credit3g

In this section, students are required to write an SQL script containing at least four mock data rows for the ***CAR5520*** table (including the mandatory sample record and user-defined records) and save the script into a file named ***Insert_car_sql.txt*** to populate the database.

*Insert_car_sql - Notepad

File Edit View

```
INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('1AMT7U', 'Aston Martin', 'Vantage', TO_DATE('March 12 2022', 'Month DD YYYY'), 5300);

INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('51F12345', 'Toyota', 'Camry Hieu', TO_DATE('January 15 2023', 'Month DD YYYY'), 15000);

INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('29A99999', 'Honda', 'Civic', TO_DATE('July 20 2021', 'Month DD YYYY'), 32000);

INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('60B88888', 'Truong Ngoc Gia Hieu', 'VF8', TO_DATE('November 05 2023', 'Month DD YYYY'), 8500);

COMMIT;
```

Figure 18: SQL Script to Insert Data into CAR5520 Table.

Enter Statements:

```
INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('99XYZ9', 'Aston Martin', 'Vantage', TO_DATE('March 12 2022', 'Month
DD YYYY'), 5300);

INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('51F99999', 'Toyota', 'Camry Hieu', TO_DATE('January 15 2023', 'Month
DD YYYY'), 15000);

INSERT INTO CAR5520 (RegNo, Make, Model, SaleDate, OdometerReading)
VALUES ('29A88888', 'Honda', 'Civic', TO_DATE('July 20 2021', 'Month DD YYYY'),
```

Execute Clear

Figure 19: Execution Result for Inserting Data into CAR5520 Table.

Notably, iSQLJunior does not allow multiple rows to be inserted in a single script execution, so each insert statement had to be run individually to populate the ***CAR5520*** table successfully.

REGNO	MAKE	MODEL	SALEDATE	ODOMETERREADING
29A88888	Honda	Civic	20-JUL-21	32000
99XYZ9	Aston Martin	Vantage	12-MAR-22	5300
51F99999	Toyota	Camry Hieu	15-JAN-23	15000
60B99999	Truong Ngoc Gia Hieu	VF8	05-NOV-23	8500

Figure 20: Data Content of CAR5520 Table.

H. Task Credit3h

In this section, students are required to write an SQL statement to list all rows of the ***CAR5520*** table in ascending order of the primary key.

```
Enter Statements:
SELECT * FROM CAR5520
ORDER BY RegNo ASC;
```

Figure 21: SQL Query Statement to Display All Records Sorted by Primary Key.

REGNO	MAKE	MODEL	SALEDATE	ODOMETERREADING
29A88888	Honda	Civic	20-JUL-21	32000
51F99999	Toyota	Camry Hieu	15-JAN-23	15000
60B99999	Truong Ngoc Gia Hieu	VF8	05-NOV-23	8500
99XYZ9	Aston Martin	Vantage	12-MAR-22	5300

OK. 4 rows selected.

Figure 22: Output Result for CAR5520 Records Sorted by Primary Key.

I. Task Credit3i

Based on the requirement, students are required to write and execute an SQL statement to list all rows of the CAR5520 table for cars that have traveled less than a specified distance.

```
Enter Statements:
SELECT * FROM CAR5520
WHERE OdometerReading < 10000
```

Figure 23: SQL Query Statement for Cars Traveled Less Than 10,000 Distance Units.

REGNO	MAKE	MODEL	SALEDATE	ODOMETERREADING
99XYZ9	Aston Martin	Vantage	12-MAR-22	5300
60B99999	Truong Ngoc Gia Hieu	VF8	05-NOV-23	8500

OK. 2 rows selected.

Figure 24: Output Result for Cars Traveled Less Than 10,000 Distance Units.

J. Task Credit3j

In the final task, students are required to write and execute an SQL statement to list all rows of the ***CAR5520*** table for cars that have a sale date after a specified date (for example, after ***January 01, 2022***).

```
Enter Statements:   SELECT * FROM CAR5520
                    WHERE SaleDate > TO_DATE('01-JAN-2022', 'DD-MON-YYYY')
```

Figure 25: SQL Query Statement for Cars with Sale Date After January 01, 2022.

REGNO	MAKE	MODEL	SALEDATE	ODOMETERREADING
99XYZ9	Aston Martin	Vantage	12-MAR-22	5300
51F99999	Toyota	Camry Hieu	15-JAN-23	15000
60B99999	Truong Ngoc Gia Hieu	VF8	05-NOV-23	8500

OK. 3 rows selected.

Figure 26: Output Result for Cars with Sale Date After January 01, 2022.

Conclusion

In conclusion, this laboratory successfully reinforced relational database management and SQL querying skills using Oracle iSQLJunior. Through practical tasks, students created, populated, and queried the CAR5520 table using primary key sorting, advanced filtering, pattern matching, and date-based conditions. Furthermore, all implemented statements and verification results demonstrate a strong understanding of database design principles and accurate SQL execution for academic reporting.